

Engineering Physics (22PH1005)- Summer Term -ACC

CO1: Home assignments:

1. (a) Define the space lattice, basis, and crystal structure.
(b) Differentiate crystalline and amorphous material. What is a unit cell? Distinguish between primitive and non-primitive unit cells.
(c) Explain the seven crystal (with neat sketch) systems with their lattice parameters (a, b, c & α, β, γ).
(d) Discuss the simple cubic (SC), body-centered cubic (BCC), and face-centered cubic (FCC) structures with clear diagrams and show how to calculate the atomic radius, coordination number, and atomic packing.
3. Explain the concept of Miller indices. (a) Derive the procedure for finding Miller indices for crystal planes and directions with examples. (b) Sketch the crystal planes for the Miller indices $(1\ 0\ 0)$, $(0\ 0\ 1)$, (101) , (111) , $(1,2,0)$, and $(1\ 2\ 1)$.
4. Explain the principle of X-ray diffraction in crystals. Explain the application of Bragg's Law in determining crystal structure. Describe the determination of the lattice constant of a given cubic crystalline material using the powder XRD method.
(a) Monochromatic X-rays of $\lambda = 1.85$ A.U. are incident on a crystal face having an inter-planar spacing of 1.26 A.U. Find the highest order for which Bragg's reflection maximum can be seen.
(b) For BCC iron, compute (a) the interplanar spacing and (b) the diffraction angle for the (110) set of planes. The lattice parameter for Fe is 0.2278nm. Also, assume that monochromatic radiation having a wavelength of 0.1460 nm is used, and the order of reflection is 1.
5. Discuss the various types of imperfections (defects) in crystals with a neat diagram. Differentiate between point, line, and surface defects with examples.